

Corrected Mathematical Guarantees for the Koopman-Enhanced Objective

Standalone revised LaTeX theorem section for K-TreeROM

1 Purpose of This Revised Section

This document is a corrected, journal-ready version of the mathematical guarantee section for the Koopman-enhanced Grassmannian regression tree objective. It incorporates the following fixes:

1. the Procrustes result is stated as a standard alignment lemma rather than a novel contribution;
2. the coordinate-invariance result explicitly covers equivariant reduced-operator estimators, including least-squares and ridge/DMD estimators;
3. the reference frame is taken to be the node basis $\hat{\Phi}_m$, which matches the local tree construction;
4. the finite-horizon rollout bound includes the missing coordinate-mismatch term $\|\hat{\Phi}_m^T \tilde{\Phi}_i - I\|$, which is absorbed into the Grassmannian term, producing a factor of two in the geometric constant;
5. the upper-bound theorem is stated for arbitrary $\omega > 0$, with a separate special case showing when the objective is exactly proportional to the bound proxy;
6. the finite-horizon Frobenius bound is explicitly limited to the aligned-Frobenius branch $d^* = d_F$. The spectral branch is still useful for frame-free regime separation, but it does not by itself control $\|\tilde{A}_i^r - \tilde{A}_m^r\|_F$.

2 Definitions

Let $\Phi_i \in \mathbb{R}^{n \times r}$ denote a local POD basis with orthonormal columns,

$$\Phi_i^T \Phi_i = I_r.$$

For a candidate node or region R_m , let $\hat{\Phi}_m \in \mathbb{R}^{n \times r}$ denote the locally aggregated POD basis, also with orthonormal columns,

$$\hat{\Phi}_m^T \hat{\Phi}_m = I_r.$$

The basis Φ_i represents a point on the Grassmann manifold $\mathcal{G}(r, n)$, and is unique only up to a right orthogonal transformation:

$$\Phi_i \sim \Phi_i S, \quad S \in \text{O}(r).$$

Consequently, a reduced Koopman operator is coordinate-dependent. If

$$A_i^r = \Phi_i^T A_i \Phi_i,$$

then under $\Phi_i \mapsto \Phi_i S$, one obtains

$$A_i^r \mapsto S^T A_i^r S.$$

Thus, Frobenius-norm comparisons between reduced Koopman operators are meaningful only after the operators are expressed in a common reduced coordinate system.

For the node R_m , set the reference basis to be $\hat{\Phi}_m$. Define

$$P_{i,m} = \Phi_i^T \hat{\Phi}_m,$$

and let

$$P_{i,m} = U_{i,m} \Sigma_{i,m} V_{i,m}^T$$

be its singular value decomposition. The Procrustes alignment from Φ_i to $\hat{\Phi}_m$ is

$$Q_{i,m}^* = U_{i,m} V_{i,m}^T.$$

The aligned local basis and aligned reduced Koopman operator are

$$\tilde{\Phi}_i = \Phi_i Q_{i,m}^*, \quad \tilde{A}_i^r = (Q_{i,m}^*)^T A_i^r Q_{i,m}^*.$$

The node Koopman operator, fitted directly in the node basis $\hat{\Phi}_m$, is denoted $\hat{A}_m^r$. Since it is already expressed in the node coordinates, write

$$\tilde{A}_m^r = \hat{A}_m^r.$$

The aligned Frobenius discrepancy is

$$d_F(i, m) = \|\tilde{A}_i^r - \tilde{A}_m^r\|_F.$$

The spectral discrepancy is

$$d_\sigma(A, B) = \min_{\pi \in S_r} \max_{1 \leq k \leq r} |\mu_k(A) - \mu_{\pi(k)}(B)|,$$

where $\{\mu_k(A)\}_{k=1}^r$ are the eigenvalues of A . The hybrid discrepancy is

$$d^*(i, m) = \begin{cases} \|\tilde{A}_i^r - \tilde{A}_m^r\|_F, & \sigma_{\min}(\Phi_i^T \hat{\Phi}_m) \geq \cos(\pi/4), \\ d_\sigma(A_i^r, \hat{A}_m^r), & \sigma_{\min}(\Phi_i^T \hat{\Phi}_m) < \cos(\pi/4). \end{cases}$$

For a candidate split $s = (j, l)$, define the Grassmannian dispersion

$$G(s) = \sum_{i \in R_L(s)} \delta(\Phi_i, \hat{\Phi}_L) + \sum_{i \in R_R(s)} \delta(\Phi_i, \hat{\Phi}_R),$$

and the Koopman dispersion

$$K(s) = \sum_{i \in R_L(s)} d^*(i, L) + \sum_{i \in R_R(s)} d^*(i, R).$$

The geometry-only TreeROM objective minimizes $G(s)$, whereas the Koopman-enhanced objective is

$$J_\omega(s) = \frac{G(s)}{\bar{\delta}_0} + \omega \frac{K(s)}{\bar{d}_0}, \quad \omega > 0.$$

Lemma 1 (Orthogonal Procrustes alignment). *Let $\Phi_i, \widehat{\Phi}_m \in \mathbb{R}^{n \times r}$ have orthonormal columns. Then*

$$Q_{i,m}^* = \arg \min_{Q \in O(r)} \|\Phi_i Q - \widehat{\Phi}_m\|_F$$

is given by

$$Q_{i,m}^* = U_{i,m} V_{i,m}^T,$$

where

$$\Phi_i^T \widehat{\Phi}_m = U_{i,m} \Sigma_{i,m} V_{i,m}^T.$$

If $\Phi_i^T \widehat{\Phi}_m$ is nonsingular and the usual Procrustes nondegeneracy conditions hold, the minimizer is unique.

Proof. Expand the objective:

$$\|\Phi_i Q - \widehat{\Phi}_m\|_F^2 = \text{tr} \left((\Phi_i Q - \widehat{\Phi}_m)^T (\Phi_i Q - \widehat{\Phi}_m) \right).$$

Since both bases are orthonormal,

$$\|\Phi_i Q - \widehat{\Phi}_m\|_F^2 = 2r - 2 \text{tr}(Q^T \Phi_i^T \widehat{\Phi}_m).$$

Thus the problem is equivalent to maximizing

$$\text{tr}(Q^T P_{i,m}), \quad P_{i,m} = \Phi_i^T \widehat{\Phi}_m.$$

Using the SVD $P_{i,m} = U_{i,m} \Sigma_{i,m} V_{i,m}^T$,

$$\text{tr}(Q^T P_{i,m}) = \text{tr}(V_{i,m}^T Q^T U_{i,m} \Sigma_{i,m}).$$

Let

$$M = V_{i,m}^T Q^T U_{i,m}.$$

Since $Q, U_{i,m}, V_{i,m} \in O(r)$, we have $M \in O(r)$. Hence

$$\text{tr}(Q^T P_{i,m}) = \sum_{k=1}^r M_{kk} \sigma_k \leq \sum_{k=1}^r \sigma_k,$$

because $|M_{kk}| \leq 1$. Equality is achieved when $M = I_r$, which gives

$$V_{i,m}^T Q^T U_{i,m} = I_r.$$

Therefore,

$$Q = U_{i,m} V_{i,m}^T.$$

□

Lemma 2 (Equivariance of reduced Koopman estimators). *Let $Z = \Phi_i^T X$ and $Z^+ = \Phi_i^T X^+$ denote reduced time-shifted data in the local coordinates. Consider the ridge-regularized reduced operator estimator*

$$A_r(Z, Z^+) = Z^+ Z^T (Z Z^T + \gamma I_r)^{-1}, \quad \gamma \geq 0,$$

with the Moore-Penrose interpretation when $\gamma = 0$. If the local basis is rotated by $S \in O(r)$, so that

$$Z' = S^T Z, \quad Z^{+'} = S^T Z^+,$$

then

$$A_r(Z', Z^{+'}) = S^T A_r(Z, Z^+) S.$$

The same equivariance holds for the projected full-order operator $A_i^r = \Phi_i^T A_i \Phi_i$.

Proof. For $\gamma > 0$,

$$A_r(Z', Z^{+'}) = S^T Z^+ Z^T S (S^T Z Z^T S + \gamma I_r)^{-1}.$$

Since S is orthogonal,

$$S^T Z Z^T S + \gamma I_r = S^T (Z Z^T + \gamma I_r) S,$$

and therefore

$$(S^T (Z Z^T + \gamma I_r) S)^{-1} = S^T (Z Z^T + \gamma I_r)^{-1} S.$$

Thus,

$$A_r(Z', Z^{+'}) = S^T Z^+ Z^T S S^T (Z Z^T + \gamma I_r)^{-1} S = S^T A_r(Z, Z^+) S.$$

For $\gamma = 0$, the same statement follows from orthogonal invariance of the Moore-Penrose pseudoinverse.

For a projected full-order operator,

$$(\Phi_i S)^T A_i (\Phi_i S) = S^T (\Phi_i^T A_i \Phi_i) S.$$

□

Theorem 1 (The aligned Frobenius discrepancy is coordinate-invariant). *Assume the reduced Koopman estimator is equivariant under right-orthogonal basis rotations, as in the previous lemma. If the local POD basis is replaced by another basis for the same subspace,*

$$\Phi'_i = \Phi_i S_i, \quad S_i \in O(r),$$

then the aligned Koopman operator is unchanged:

$$\tilde{A}_i^{r'} = \tilde{A}_i^r.$$

Consequently,

$$\|\tilde{A}_i^r - \tilde{A}_m^r\|_F$$

is independent of arbitrary POD basis orientations.

Proof. Under the basis rotation $\Phi'_i = \Phi_i S_i$, equivariance gives

$$A_i^{r'} = S_i^T A_i^r S_i.$$

Moreover,

$$(\Phi'_i)^T \hat{\Phi}_m = S_i^T \Phi_i^T \hat{\Phi}_m.$$

If

$$\Phi_i^T \hat{\Phi}_m = U_{i,m} \Sigma_{i,m} V_{i,m}^T,$$

then

$$(\Phi'_i)^T \hat{\Phi}_m = (S_i^T U_{i,m}) \Sigma_{i,m} V_{i,m}^T.$$

Hence the new Procrustes factor is

$$Q'_{i,m} = S_i^T U_{i,m} V_{i,m}^T = S_i^T Q_{i,m}^*.$$

Therefore,

$$\begin{aligned}
\tilde{A}_i^{r'} &= (Q'_{i,m})^T A_i^{r'} Q'_{i,m} \\
&= (S_i^T Q_{i,m}^*)^T (S_i^T A_i^r S_i) (S_i^T Q_{i,m}^*) \\
&= (Q_{i,m}^*)^T S_i S_i^T A_i^r S_i S_i^T Q_{i,m}^* \\
&= (Q_{i,m}^*)^T A_i^r Q_{i,m}^* \\
&= \tilde{A}_i^r.
\end{aligned}$$

Thus the aligned operator depends only on the subspace and the node reference frame, not on the arbitrary numerical POD basis returned by the SVD. $\square$

Proposition 1 (The $\pi/4$ alignment certificate is the exact closeness boundary). *Let $\theta_{\max}$ be the largest principal angle between*

$$\text{span}(\Phi_i) \quad \text{and} \quad \text{span}(\hat{\Phi}_m).$$

Then

$$\sigma_{\min}(\Phi_i^T \hat{\Phi}_m) = \cos \theta_{\max}.$$

Moreover,

$$\theta_{\max} \leq \frac{\pi}{4}$$

if and only if every aligned principal direction is at least as close to the node subspace as to its orthogonal complement. Equivalently,

$$\sigma_{\min}(\Phi_i^T \hat{\Phi}_m) \geq \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}.$$

Proof. Let $u \in \text{span}(\Phi_i)$ be a principal direction with principal angle θ . Then

$$u = \cos \theta v + \sin \theta w,$$

where

$$v \in \text{span}(\hat{\Phi}_m), \quad w \in \text{span}(\hat{\Phi}_m)^\perp, \quad \|v\| = \|w\| = 1.$$

The distance from u to the node subspace is

$$\text{dist}(u, \text{span}(\hat{\Phi}_m)) = \sin \theta,$$

whereas the distance from u to the orthogonal complement is

$$\text{dist}(u, \text{span}(\hat{\Phi}_m)^\perp) = \cos \theta.$$

Thus u is at least as close to the node subspace as to its orthogonal complement if and only if

$$\sin \theta \leq \cos \theta.$$

This is equivalent to

$$\theta \leq \frac{\pi}{4}.$$

Applying this to every principal direction gives

$$\theta_{\max} \leq \frac{\pi}{4}.$$

Finally, the singular values of $\Phi_i^T \widehat{\Phi}_m$ are

$$\sigma_k(\Phi_i^T \widehat{\Phi}_m) = \cos \theta_k.$$

Therefore,

$$\sigma_{\min}(\Phi_i^T \widehat{\Phi}_m) = \cos \theta_{\max}.$$

The result follows. $\square$

Assumption 1 (Idealized local rollout setting). *For the finite-horizon bound below, assume the following idealized setting.*

1. *The sample trajectory is exactly represented in the aligned local basis:*

$$x_k^{(i)} = \widetilde{\Phi}_i z_k, \quad \widetilde{\Phi}_i = \Phi_i Q_{i,m}^*.$$

2. *The local and node reduced rollouts use the same initial reduced coordinate:*

$$z_0 = \hat{z}_0, \quad \|z_0\| \leq M.$$

3. *The aligned local and node reduced operators are power-bounded by*

$$\|\widetilde{A}_i^r\|_2 \leq \rho, \quad \|\widetilde{A}_m^r\|_2 \leq \rho.$$

4. *The alignment certificate holds for the sample relative to the node basis:*

$$\sigma_{\min}(\Phi_i^T \widehat{\Phi}_m) \geq \cos(\pi/4).$$

Therefore, the discrepancy branch is $d^* = d_F$.

Theorem 2 (Finite-horizon rollout-error upper bound). *Under the idealized local rollout assumptions, let*

$$z_{k+1} = \widetilde{A}_i^r z_k, \quad \hat{z}_{k+1} = \widetilde{A}_m^r \hat{z}_k.$$

Then, for every $k \geq 1$,

$$\|x_k^{(i)} - \hat{x}_k^{(m)}\| \leq 2M\rho^k \delta(\Phi_i, \widehat{\Phi}_m) + Mk\rho^{k-1} \|\widetilde{A}_i^r - \widetilde{A}_m^r\|_F,$$

where

$$\hat{x}_k^{(m)} = \widehat{\Phi}_m \hat{z}_k.$$

Consequently, over a finite horizon T ,

$$\mathcal{E}_T^{(i,m)} \leq C_\Phi \delta(\Phi_i, \widehat{\Phi}_m) + C_A \|\widetilde{A}_i^r - \widetilde{A}_m^r\|_F,$$

with

$$C_\Phi = 2M \sum_{k=0}^T \rho^k, \quad C_A = M \sum_{k=1}^T k\rho^{k-1}.$$

Proof. The aligned local trajectory and the node trajectory are

$$z_k = (\tilde{A}_i^r)^k z_0, \quad \hat{z}_k = (\tilde{A}_m^r)^k z_0.$$

Let

$$\hat{\Pi}_m = \hat{\Phi}_m \hat{\Phi}_m^T, \quad C_{i,m} = \hat{\Phi}_m^T \tilde{\Phi}_i.$$

Since $x_k^{(i)} = \tilde{\Phi}_i z_k$ and $\hat{x}_k^{(m)} = \hat{\Phi}_m \hat{z}_k$, decompose the state error as

$$\begin{aligned} x_k^{(i)} - \hat{x}_k^{(m)} &= \tilde{\Phi}_i z_k - \hat{\Phi}_m \hat{z}_k \\ &= (I - \hat{\Pi}_m) \tilde{\Phi}_i z_k + \hat{\Phi}_m (\hat{\Phi}_m^T \tilde{\Phi}_i z_k - \hat{z}_k) \\ &= (I - \hat{\Pi}_m) \tilde{\Phi}_i z_k + \hat{\Phi}_m ((C_{i,m} - I) z_k + (z_k - \hat{z}_k)). \end{aligned}$$

Taking norms and using $\|\hat{\Phi}_m\|_2 = 1$,

$$\|x_k^{(i)} - \hat{x}_k^{(m)}\| \leq \|(I - \hat{\Pi}_m) \tilde{\Phi}_i z_k\| + \|(C_{i,m} - I) z_k\| + \|z_k - \hat{z}_k\|.$$

First, the orthogonal projection term is controlled by the largest principal angle:

$$\|(I - \hat{\Pi}_m) \tilde{\Phi}_i z_k\| \leq \sin \theta_{\max}(\Phi_i, \hat{\Phi}_m) \|z_k\|.$$

Since $\|z_k\| \leq M\rho^k$ and

$$\sin \theta_{\max}(\Phi_i, \hat{\Phi}_m) \leq \left(\sum_{\ell=1}^r \theta_\ell^2 \right)^{1/2} = \delta(\Phi_i, \hat{\Phi}_m),$$

we have

$$\|(I - \hat{\Pi}_m) \tilde{\Phi}_i z_k\| \leq M\rho^k \delta(\Phi_i, \hat{\Phi}_m).$$

Second, the coordinate mismatch term is not zero in general. The Procrustes alignment implies

$$C_{i,m} = \hat{\Phi}_m^T \Phi_i Q_{i,m}^* = V_{i,m} \Sigma_{i,m} V_{i,m}^T.$$

Therefore,

$$\|C_{i,m} - I\|_2 = \max_{\ell} |\cos \theta_\ell - 1| = 1 - \cos \theta_{\max}.$$

For $0 \leq \theta \leq \pi/2$, one has $1 - \cos \theta \leq \sin \theta$. Hence,

$$\|C_{i,m} - I\|_2 \leq \sin \theta_{\max} \leq \delta(\Phi_i, \hat{\Phi}_m).$$

Thus,

$$\|(C_{i,m} - I) z_k\| \leq M\rho^k \delta(\Phi_i, \hat{\Phi}_m).$$

This is the additional term that must be included; it is absorbed into the Grassmannian contribution, doubling the geometric constant.

Third, the dynamical term satisfies

$$\|z_k - \hat{z}_k\| = \left\| \left[(\tilde{A}_i^r)^k - (\tilde{A}_m^r)^k \right] z_0 \right\|.$$

Using

$$A^k - B^k = \sum_{q=0}^{k-1} A^{k-1-q} (A - B) B^q,$$

we obtain

$$\begin{aligned}
\left\| (\tilde{A}_i^r)^k - (\tilde{A}_m^r)^k \right\|_2 &\leq \sum_{q=0}^{k-1} \|\tilde{A}_i^r\|_2^{k-1-q} \|\tilde{A}_i^r - \tilde{A}_m^r\|_2 \|\tilde{A}_m^r\|_2^q \\
&\leq k\rho^{k-1} \|\tilde{A}_i^r - \tilde{A}_m^r\|_2 \\
&\leq k\rho^{k-1} \|\tilde{A}_i^r - \tilde{A}_m^r\|_F.
\end{aligned}$$

Therefore,

$$\|z_k - \hat{z}_k\| \leq Mk\rho^{k-1} \|\tilde{A}_i^r - \tilde{A}_m^r\|_F.$$

Combining the three terms gives

$$\|x_k^{(i)} - \hat{x}_k^{(m)}\| \leq 2M\rho^k \delta(\Phi_i, \hat{\Phi}_m) + Mk\rho^{k-1} \|\tilde{A}_i^r - \tilde{A}_m^r\|_F.$$

Summing over $k = 0, \dots, T$ gives the finite-horizon result with

$$C_\Phi = 2M \sum_{k=0}^T \rho^k, \quad C_A = M \sum_{k=1}^T k\rho^{k-1}.$$

□

Remark 1 (Scope of the Frobenius rollout bound). *The previous theorem uses $\|\tilde{A}_i^r - \tilde{A}_m^r\|_F$. Therefore, it applies directly on the aligned-Frobenius branch $d^\star = d_F$, i.e., when the alignment certificate is satisfied. The spectral branch d_σ is frame-free and useful for detecting dynamically distinct regimes, but d_σ alone does not control the Frobenius-norm difference of nonnormal operators. Thus, the finite-horizon Frobenius rollout bound should be stated only for node/sample pairs passing the alignment certificate.*

Theorem 3 (K-TreeROM gives a bound-controlled objective for any $\omega > 0$). *Assume all sample/node pairs involved in a candidate split satisfy the aligned-Frobenius certificate, so that $d^\star = d_F$. Then the finite-horizon split error satisfies*

$$\mathcal{E}_T(s) \leq C_\Phi G(s) + C_A K(s).$$

For any $\omega > 0$, define

$$C_\omega = \max \left\{ C_\Phi \bar{\delta}_0, \frac{C_A \bar{d}_0}{\omega} \right\}.$$

Then

$$\mathcal{E}_T(s) \leq C_\omega J_\omega(s).$$

Thus, every positive value of ω gives a valid scaled upper-bound proxy. Moreover, if

$$\omega^{star} = \frac{C_A \bar{d}_0}{C_\Phi \bar{\delta}_0},$$

then

$$J_{\omega^{star}}(s) = \frac{1}{C_\Phi \bar{\delta}_0} [C_\Phi G(s) + C_A K(s)],$$

so minimizing $J_{\omega^{star}}$ is exactly equivalent to minimizing the finite-horizon bound proxy

$$B(s) = C_\Phi G(s) + C_A K(s).$$

Proof. From the previous theorem summed over all samples assigned by split s ,

$$\mathcal{E}_T(s) \leq C_\Phi G(s) + C_A K(s).$$

Rewrite

$$G(s) = \bar{\delta}_0 \frac{G(s)}{\bar{\delta}_0}, \quad K(s) = \frac{\bar{d}_0}{\omega} \left(\omega \frac{K(s)}{\bar{d}_0} \right).$$

Therefore,

$$\begin{aligned} C_\Phi G(s) + C_A K(s) &= C_\Phi \bar{\delta}_0 \frac{G(s)}{\bar{\delta}_0} + \frac{C_A \bar{d}_0}{\omega} \left(\omega \frac{K(s)}{\bar{d}_0} \right) \\ &\leq \max \left\{ C_\Phi \bar{\delta}_0, \frac{C_A \bar{d}_0}{\omega} \right\} \left[\frac{G(s)}{\bar{\delta}_0} + \omega \frac{K(s)}{\bar{d}_0} \right] \\ &= C_\omega J_\omega(s). \end{aligned}$$

Thus,

$$\mathcal{E}_T(s) \leq C_\omega J_\omega(s).$$

For the special choice

$$\omega^\star = \frac{C_A \bar{d}_0}{C_\Phi \bar{\delta}_0},$$

we get

$$J_{\omega^\star}(s) = \frac{G(s)}{\bar{\delta}_0} + \frac{C_A \bar{d}_0}{C_\Phi \bar{\delta}_0} \frac{K(s)}{\bar{d}_0} = \frac{1}{C_\Phi \bar{\delta}_0} [C_\Phi G(s) + C_A K(s)].$$

Since the scaling constant is positive, minimizing $J_{\omega^\star}$ is equivalent to minimizing $B(s) = C_\Phi G(s) + C_A K(s)$. $\square$

Corollary 1 (Comparison with the geometry-only objective). *The geometry-only objective controls only $G(s)$. The Koopman-enhanced objective controls both $G(s)$ and $K(s)$. Therefore, whenever operator mismatch contributes to rollout error and $K(s)$ varies across candidate splits, the Koopman-enhanced objective is a strictly more informative error proxy than the geometry-only objective.*

Remark 2 (How to phrase the result without overclaiming). *The preceding theorem should not be read as saying that the experimental value of ω is necessarily the exact optimal theoretical value $\omega^\star$. Rather, it shows that for any $\omega > 0$, the normalized hybrid objective upper-bounds the same two physical contributors to rollout error: spatial subspace mismatch and reduced-operator mismatch. The special value $\omega^\star$ only gives the exact proportionality between the objective and the idealized bound proxy.*

Proposition 2 (Geometry-only TreeROM can be non-identifiable). *Assume an idealized two-regime system with identical subspaces but different dynamics:*

$$\Phi(\lambda) \equiv \Phi,$$

and

$$A(\lambda) = \begin{cases} A_1, & \lambda \leq \lambda_s, \\ A_2, & \lambda > \lambda_s. \end{cases}$$

Then the geometry-only objective is flat:

$$G(s) = 0 \quad \text{for all candidate splits } s.$$

Thus, the true dynamical regime boundary is not identifiable from Grassmannian dispersion alone.

Proof. Since $\Phi_i = \Phi$ for every sample, the locally aggregated basis in any region is also $\hat{\Phi}_m = \Phi$, in the idealized noise-free setting. Hence

$$\delta(\Phi_i, \hat{\Phi}_m) = 0$$

for every sample and every region. Therefore,

$$G(s) = 0$$

for all candidate splits s . The geometry-only objective has no separating signal for the true boundary. $\square$

Proposition 3 (The Koopman term restores a positive separation signal). *In the same two-regime setting, assume the operator spectra are separated by*

$$g = d_\sigma(A_1, A_2) > 0.$$

For any mixed region with representative $\hat{A}$, the triangle inequality gives

$$\max \left\{ d_\sigma(A_1, \hat{A}), d_\sigma(A_2, \hat{A}) \right\} \geq \frac{g}{2}.$$

Thus, any split that mixes spectrally distinct regimes pays a positive Koopman penalty, whereas a pure split does not in the idealized noise-free case.

Proof. By the triangle inequality for the bottleneck spectral matching distance,

$$d_\sigma(A_1, A_2) \leq d_\sigma(A_1, \hat{A}) + d_\sigma(\hat{A}, A_2).$$

Since $d_\sigma(A_1, A_2) = g$,

$$g \leq d_\sigma(A_1, \hat{A}) + d_\sigma(A_2, \hat{A}).$$

Therefore, at least one of the two terms must be at least $g/2$:

$$\max \left\{ d_\sigma(A_1, \hat{A}), d_\sigma(A_2, \hat{A}) \right\} \geq \frac{g}{2}.$$

This gives a positive dynamical separation signal on mixed regions. $\square$

3 Journal-Ready Summary Paragraph

The theoretical results above justify the Koopman-enhanced splitting criterion as follows. The Procrustes alignment places locally computed reduced operators into the node coordinate frame, and the resulting aligned Frobenius discrepancy is invariant to arbitrary right-orthogonal rotations of the POD basis. The certificate $\sigma_{\min}(\Phi_i^T \hat{\Phi}_m) \geq \cos(\pi/4)$ is the exact condition under which every aligned principal direction remains at least as close to the node subspace as to its orthogonal complement. Under this certificate and bounded reduced dynamics, the finite-horizon rollout error is bounded by a spatial term proportional to $\delta(\Phi_i, \hat{\Phi}_m)$ and a dynamical term proportional to $\|\tilde{A}_i^r - \tilde{A}_m^r\|_F$. Hence the proposed objective controls both subspace projection mismatch and accumulated reduced-dynamics mismatch, while the original Grassmannian objective controls only the spatial component. When subspaces are identical or nearly identical but the reduced dynamics differ, the geometry-only objective can be non-identifiable, whereas the Koopman term provides a positive separation signal proportional to the spectral gap.

References

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